

Quiz 1 (Version 1)

CS 320: Concepts of Programming Languages

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- You will have approximately 30 minutes to complete this quiz. Make sure to read every question; some are easier than others.
- Make very clear what your final solution for each problem is (e.g., by surrounding it in a box). We reserve the right to mark off points if we cannot tell what your final solution is.
- You must show your work on all problems unless otherwise specified. A solution without work will be considered incorrect (and will be investigated for potential academic misconduct).
- We will not look at any work on the pages marked *"This page is intentionally left blank."*. You should use these pages for scratch work.

1 OCaml Programming

(10 points) Implement the function

```
val range : int -> int -> int list
```

so that `range i j` is the list of integers from `i` to `j` inclusive in increasing order. If `i > j` then the range is empty.

```
let _ = assert (range 3 6 = [3;4;5;6])
let _ = assert (range 6 3 = [])
```

You may not use any functions from the standard library except for arithmetic operators and list constructors (i.e., `[]` and `::`).

let rec range i j = if i > j then [] else i :: range (i+1) j

2 Types and Values

(10 points) Determine the type of the function `foo` and the value of `bar` below.

```
let rec foo l =  
  match l with  
  | [] -> 0  
  | [] :: xs -> foo xs - 1  
  | (y :: ys) :: xs -> y + foo (ys :: xs)  
let bar = foo [[1;2;3];[4;5];[6]]
```

Justify your answers.

`val foo : int list list → int`

$$\begin{aligned} \text{foo } [[1;2;3]; [4;5]; [6]] &= \\ 1 + \text{foo } [[2;3]; [4;5]; [6]] &= \\ 3 + \text{foo } [[3]; [4;5]; [6]] &= \\ 6 + \text{foo } [[]; [4;5]; [6]] &= \\ 6 + (\text{foo } [[4;5]; [6]] - 1) &= \\ 6 + (4 + \text{foo } [[5]; [6]] - 1) &= \\ 6 + (9 + \text{foo } [[]; [6]] - 1) &= \\ 6 + (9 + (\text{foo } [6] - 1)) - 1 &= \\ 6 + (9 + (6 + \text{foo } [] - 1) - 1) &= \\ 6 + (9 + (6 + (0 - 1) - 1) - 1) &= \\ 18 \end{aligned}$$

Note: This function finds the sum of the ints minus the number of inner lists

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